

Decision Journal — Week 7 Model Choice

Date: 2026-07-21 Decision owner: Mithra Ramoutar

Context

- The Week 6 baseline (logistic regression) passed Dr. Reyes' review, but the ED Board and Martina Griffith (Clinical IT) asked whether a more complex model is worth its extra cost — in accuracy, compute, and explainability.
- Recall on ESI 1 was already established in Week 6 as the primary clinical metric, so any new model has to be judged against that number specifically, not against overall accuracy.

Alternatives considered

- **Random Forest** (untuned and hyperparameter-tuned via `RandomizedSearchCV`) — expected to be a strong, moderately interpretable option per the literature; in practice it underperformed the baseline on every quality metric, including 0% recall on ESI 1 untuned.
- **Small MLP (neural network)** — flexible, but per the tutorial's own guidance, rarely beats tree models on tabular clinical data and is hard to explain; confirmed here — no better than the baseline on macro F1 and no better on ESI 1 recall.
- **Gradient Boosting** — the only model that improved on ESI 1 recall (0.313 vs. baseline's 0.250), though at a cost in accuracy, precision, and explainability.

Decision

Carry Gradient Boosting into Phase 3 piloting, paired with mandatory SHAP explanations per prediction, while keeping the Week 6 logistic regression baseline as a documented fallback.

Reasoning

- Gradient Boosting is the only tested model that improves the metric we already told the ED Board matters most — recall on ESI 1 — rather than trading it away for a better-looking aggregate number.
- Its inference cost (0.009 ms/prediction) is negligible regardless of which model is chosen, so compute cost does not argue against it the way it might for a larger/heavier model.
- Explainability is a real cost, not a free pass — SHAP is being adopted as a permanent operational requirement alongside the model, not treated as optional polish.

Things I do not yet know

- Whether the ESI 1 recall gain (one additional patient caught, out of 16 in the test set) holds up under cross-validation or a larger sample — the current result rests on a very small absolute count and could easily be noise.
- How this recommendation changes once validated against real Mercer General data rather than the US academic-hospital benchmark dataset, per the standing caveat from the Week 5 memo.